Shaden **Shaar**

NLP Researcher · PhD Candidate at Cornell University

Ithaca, NY, USA

ss2753 [at] cornell [dot] edu | sshaar.github.io | sshaar | Google Scholar

About Me

I am a PhD candidate at Cornell University, advised by Prof. Claire Cardie, working on *Long-Form Input Understanding across Vision-and-Language Tasks*. I have worked across several domains and modalities: document-level event extraction in text (TACL 2025), and open-ended question answering over movie recaps with *MovieRecapsQA* (CVPR 2026) and now over full-length films. During my PhD, I have also collaborated with NewYork-Presbyterian Hospital on cardiology NLP projects, including an LLM-based analysis of UNOS exception requests (JHLT 2026). Before Cornell, I spent two years as a research assistant at the Qatar Computing Research Institute working on fact-checking and propaganda detection across modalities. I did my undergrad in Computer Science at Carnegie Mellon University.

Research Interests

- **Long-form input understanding.** How models process long documents, short videos, and full-length films, and what each task requires them to locate in the input.
- **Benchmarks and evaluation for open-ended generation.** Building datasets on which current models fail, and scoring long, open-ended answers against rubrics rather than a single reference answer.
- **Clinical NLP.** Using LLMs to identify clinical decision patterns in unstructured medical narratives, primarily in heart failure and heart transplant care.
- **Fact-checking and propaganda detection** *(prior work).* Detecting previously fact-checked claims, and detecting propaganda and persuasion techniques in text, images, and memes.

Highlights

- **h-index 24** with ∼3,100 citations; 32 publications across CVPR, TACL, ACL, EMNLP, NAACL, COLING, IJCAI, ICWSM, and LREC.
- Visiting the **Berkeley NLP group**, UC Berkeley (Fall 2026), working with Dr. Sewon Min.
- **University Fellowship**, Cornell University (2021).
- **Co-organized shared tasks** at CLEF–CheckThat! (2020, 2021), SemEval-2021 Task 6 on persuasion techniques in memes, and NLP4IF-2021.
- **Introduced the task** of detecting previously fact-checked claims (ACL 2020, 240+ citations), now a recurring task at the CLEF–CheckThat! lab.
- **Best Demo Award (Honorable Mention)**, ACL 2020, for *Prta*, a system for analyzing propaganda techniques in news.

Education

Ph.D. in Computer Science — *Ithaca, NY, USA*

Cornell University — *Aug. 2021 – Jan. 2027 (expected)*

Minor in Applied Mathematics. Advised by Prof. Claire Cardie.

M.S. in Computer Science — *Ithaca, NY, USA*

Cornell University — *Aug. 2021 – May 2024*

B.S. in Computer Science — *Doha, Qatar*

Carnegie Mellon University — *Aug. 2015 – May 2019*

Minor in Mathematics. College and University Honors.

Work Experience

Zillow Group — *Remote, USA*

Applied Scientist Intern — *May 2025 – Aug. 2025*

- Built a home-purchase assistant: a multi-turn chatbot that answers buyer questions, tracks purchase intent over the conversation, and narrows a large listing inventory to matching properties.
- Post-trained the dialogue model on long multi-turn conversations, first with DPO on preference pairs over whole dialogues, then with GRPO using a trajectory-level reward rather than a per-turn reward.
- Built the reward model and the evaluation pipeline for this training, including rollout generation and conversation-level scoring for grounded property-search dialogue.

ScaleAI, Inc. *New York City, NY, USA*

Machine Learning Research Engineer Intern *Jan. 2025 – May 2025*

- Built precedent retrieval as a hybrid RAG system over constitutional bylaws and prior Qatari Supreme Court cassation rulings: dense embedding retrieval fused with sparse lexical matching, followed by reranking of the merged candidate set.
- Fine-tuned an Arabic LLM with SFT on data collected specifically for Qatari court cases, for event extraction and case summarization.
- Designed the Arabic information-extraction pipeline: OCR over scanned filings, then event extraction that links each event to its supporting evidence spans, feeding a structured case summarizer.
- Deployed the above as API services behind a judge-facing case-analysis dashboard for the Qatari judicial department.

Apple *Seattle, WA, USA*

AI/ML Research Intern *May 2022 – Aug. 2022*

- Built a conditional synthetic dialogue generator for Siri training data, targeting scenarios with little or no real usage data. Mentored by Dr. Alex Churchill.
- Simulated the user side of each conversation so the pipeline produced complete multi-turn dialogues rather than isolated utterances.
- Conditioned generation on intent and user behavior, allowing a target dataset to be specified by requirements, and ran it zero-shot without seed dialogues.

Qatar Computing Research Institute *Doha, Qatar*

Research Assistant *Jul. 2019 – Aug. 2021*

- Research assistant advised by Dr. Preslav Nakov and Prof. Giovanni Da San Martino.
- Introduced the task of detecting previously fact-checked claims, with a retrieve-then-rerank pipeline: BM25 candidate retrieval, Sentence-BERT semantic similarity, and a learning-to-rank reranker (ACL 2020). Extended it to document-level claim retrieval and to context modeling with Transformer encoders over surrounding tweets and co-mentions (Findings of EMNLP 2022, Findings of NAACL 2022, IJCAI 2021).
- Built *Prta*, a public system for fine-grained propaganda detection in news articles, combining span-level sequence tagging and sentence-level classification with a BERT-based multi-granularity network (ACL 2020, Best Demo Award Honorable Mention). Extended it to memes as multi-modal multi-label classification with text–image fusion models (ACL 2021, SemEval-2021).
- Built multilingual, multi-label classifiers on fine-tuned Transformer encoders for the factuality, harmfulness, and framing of COVID-19 claims across Arabic, English, and Bulgarian social media (ICWSM 2021, Findings of EMNLP 2021, RANLP 2021, COLING 2022).
- Built cross-lingual emotion classifiers for low-resource languages using multilingual encoders and machine translation for transfer (LREC 2022).
- Co-organized shared tasks on check-worthiness estimation and verified-claim retrieval at CLEF–CheckThat! 2020 and 2021, on persuasion techniques in memes at SemEval-2021 Task 6, and at NLP4IF-2021, releasing the datasets and evaluation setups.

Publications

Under Review

1. **Shaar, S.**, Thymes, B., Aldaye, M., Cardie, C., and Hariharan, B. "LF-VideoQA: Evaluating Long-Video Understanding with Long-Form Theory-of-Mind Questions." Under review, 2026.

Journals

1. Frye, J., **Shaar, S.**, Cardie, C., DeFilippis, E., Estrin, D., Sayer, G., Uriel, N., et al. "Thematic Analysis of Accepted Exception Requests for Heart Transplant Candidates Using a Large Language Model." *The Journal of Heart and Lung Transplantation* 45(5):460, 2026. **(JHLT'2026)**.
2. **Shaar, S.**, Chen, W., Chatterjee, M., Wang, B., Zhao, W., and Cardie, C. "Are *Triggers* Needed for Document-Level Event Extraction?" *Transactions of the Association for Computational Linguistics* **(TACL'2025)**.

Conferences

1. **Shaar, S.**, Thymes, B., Chaixanien, S., Cardie, C., and Hariharan, B. "MovieRecapsQA: A Multimodal Open-Ended Video Question-Answering Benchmark." *Conference on Computer Vision and Pattern Recognition* **(CVPR'2026)**.
2. Chaixanien, S., Choi, E., **Shaar, S.**, and Cardie, C. "Pungene at DialAM-2024: Identification of Propositional and Illocutionary Relations." *Workshop on Argument Mining at ACL 2024* **(ArgumentMining@ACL'2024)**.
3. Aydin, A. N., **Shaar, S.**, and Cardie, C. "Edward Said at Touché: Human Value Detection Using Transformers and Upsampling." *Working Notes of the Conference and Labs of the Evaluation Forum* **(CLEF'2024)**.
4. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "The CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *European Conference on Information Retrieval* **(ECIR'2022)**.
5. Nakov, P., Barrón-Cedeño, A., Da San Martino, G., Alam, F., Struß, J. M., Mandl, T., Míguez, R., Caselli, T., Kutlu, M., Zaghouani, W., et al. "Overview of the CLEF-2022 CheckThat! Lab on Fighting the COVID-19 Infodemic and Fake News Detection." *Cross-Language Evaluation Forum for European Languages* **(CLEF'2022)**.

6. **Shaar, S.**, Georgiev, N., Alam, F., Da San Martino, G., Mohamed, A., and Nakov, P. "Assisting the Human Fact-Checkers: Detecting All Previously Fact-Checked Claims in a Document." *Findings of the Association for Computational Linguistics: EMNLP 2022* **(EMNLP'2022)**.
7. **Shaar, S.**, Alam, F., Da San Martino, G., and Nakov, P. "The Role of Context in Detecting Previously Fact-Checked Claims." *Findings of the Association for Computational Linguistics: NAACL 2022* **(NAACL'2022)**.
8. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "A Second Pandemic? Analysis of Fake News About COVID-19 Vaccines in Qatar." *Findings of the Association for Computational Linguistics: NAACL 2022* **(NAACL'2022)**.
9. Hassan, S., **Shaar, S.**, and Darwish, K. "Cross-Lingual Emotion Detection." *Proceedings of the Thirteenth Language Resources and Evaluation Conference* **(LREC'2022)**.
10. Alam, F., Cresci, S., Chakraborty, T., Silvestri, F., Dimitrov, D., Da San Martino, G., **Shaar, S.**, Firooz, H., and Nakov, P. "A Survey on Multimodal Disinformation Detection." *International Conference on Computational Linguistics* **(COLING'2022)**.
11. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "Detecting Propaganda Techniques in Memes." *Association for Computational Linguistics* **(ACL–IJCNLP'2021)**.
12. Dimitrov, D., Bin Ali, B., **Shaar, S.**, Alam, F., Silvestri, F., Firooz, H., Nakov, P., and Da San Martino, G. "SemEval-2021 Task 6: Detection of Persuasion Techniques in Texts and Images." *ACL-IJCNLP SemEval Workshop* **(SemEval@ACL-IJCNLP'2021)**.
13. **Shaar, S.**, Alam, F., Da San Martino, G., Nikolov, A., Zaghouani, W., Nakov, P., and Feldman, A. "Findings of the NLP4IF-2021 Shared Tasks on Fighting the COVID-19 Infodemic and Censorship Detection." *ACL NLP4IF Workshop* **(NLP4IF@ACL'2021)**.
14. Nakov, P., Da San Martino, G., Elsayed, T., Barrón-Cedeño, A., Míguez, R., **Shaar, S.**, Alam, F., Haouari, F., Hasanain, M., Babulkov, N., Nikolov, A., Shahi, G., Struß, J., and Mandl, T. "The CLEF-2021 CheckThat! Lab on Detecting Check-Worthy Claims, Previously Fact-Checked Claims, and Fake News." *European Conference on Information Retrieval* **(ECIR'2021)**.
15. Skuczynska, B., **Shaar, S.**, Spenader, J., and Nakov, P. "BeaSku at CheckThat! 2021: Fine-Tuning Sentence BERT with Triplet Loss and Limited Data." *CLEF Conference and Labs of the Evaluation Forum* 639–647, **(CLEF'2021)**.
16. Alam, F., **Shaar, S.**, Dalvi, F., Sajjad, H., Nikolov, A., Mubarak, H., Da San Martino, G., Abdelali, A., Durrani, N., Darwish, K., Al-Homaid, A., Zaghouani, W., Caselli, T., Danoe, G., Stolk, F., Bruntink, F., and Nakov, P. "Fighting the COVID-19 Infodemic: Modeling the Perspective of Journalists, Fact-Checkers, Social Media Platforms, Policy Makers, and the Society." *Findings of EMNLP* **(EMNLP'2021)**.
17. Alam, F., Dalvi, F., **Shaar, S.**, Durrani, N., Mubarak, H., Nikolov, A., Da San Martino, G., Abdelali, A., Sajjad, H., Darwish, K., and Nakov, P. "Fighting the COVID-19 Infodemic in Social Media: A Holistic Perspective and a Call to Arms." *International AAAI Conference on Web and Social Media* **(ICWSM'2021)**.
18. Nakov, P., Alam, F., **Shaar, S.**, Da San Martino, G., and Zhang, Y. "COVID-19 in Bulgarian Social Media: Factuality, Harmfulness, Propaganda, and Framing." *International Conference on Recent Advances in Natural Language Processing* **(RANLP'2021)**.
19. Nakov, P., Corney, D., Hasanain, M., Alam, F., Elsayed, T., Barrón-Cedeño, A., Papotti, P., **Shaar, S.**, and Da San Martino, G. "Automated Fact-Checking for Assisting Human Fact-Checkers." *International Joint Conferences on Artificial Intelligence* **(IJCAI'2021)**.
20. **Shaar, S.**, Da San Martino, G., Babulkov, N., and Nakov, P. "That Is a Known Lie: Detecting Previously Fact-Checked Claims." *Association for Computational Linguistics* **(ACL'2020)**.
21. Da San Martino, G., **Shaar, S.**, Zhang, Y., Yu, S., Barrón-Cedeño, A., and Nakov, P. "Prta: A System to Support the Analysis of Propaganda Techniques in the News." *Association for Computational Linguistics* **(ACL'2020)**. **Best Demo Award (Honorable Mention).**
22. **Shaar, S.**, Nikolov, A., Babulkov, N., Alam, F., Barrón-Cedeño, A., Elsayed, T., Hasanain, M., Suwaileh, R., Haouari, F., Da San Martino, G., and Nakov, P. "Overview of CheckThat! 2020 English: Automatic Identification and Verification of Claims in Social Media." *CLEF Working Notes* 2696, **(CLEF'2020)**.
23. Barrón-Cedeño, A., Elsayed, T., Nakov, P., Da San Martino, G., Hasanain, M., Suwaileh, R., Haouari, F., Babulkov, N., Hamdan, B., Nikolov, A., **Shaar, S.**, and Ali, Z. S. "Overview of CheckThat 2020: Automatic Identification and Verification of Claims in Social Media." **(CLEF'2020)**, 11:215–236.
24. **Shaar, S.**, Razak, S., Dalvi, F., and Moosavi, S. "Group Identification in Crowded Environments Using Proximity Sensing." *IEEE Conference on Local Computer Networks* **(LCN'2018)**.
25. Hassan, S., **Shaar, S.**, Raj, B., and Razak, S. "Interactive Evaluation of Classifiers Under Limited Resources." *IEEE International Conference on Machine Learning and Applications* **(ICMLA'2018)**.

Honors & Awards

2021	**University Fellowship,** Cornell University, fellowship for incoming PhD students	*Ithaca, NY, USA*
2020	**Best Demo Award (Honorable Mention),** ACL 2020, for *Prta: A System to Support the Analysis of Propaganda Techniques*	*Online*
2019	**College and University Honors,** Carnegie Mellon University, for outstanding cumulative GPA	*Doha, Qatar*
2019	**Dean's List,** Carnegie Mellon University, F15, S16, F16, F17	*Doha, Qatar*
2015	**50% Academic Merit Scholarship,** Carnegie Mellon University, merit scholarship for incoming students	*Doha, Qatar*

Academic Services

Since 2020 **Reviewer,** ACL, EMNLP, NAACL, EACL, COLING, AAAI, CIKM (Program Committee)

2021	**Co-organizer,** SemEval–2021 Task 6: Detection of Persuasion Techniques in Texts and Images	*Online*
2021	**Co-organizer,** CLEF–2021 CheckThat! Lab on detecting check-worthy and previously fact-checked claims	*Bucharest, Romania*
2020	**Co-organizer,** CLEF–2020 CheckThat! Lab on automatic identification and verification of claims	*Online*

Teaching Experience

Cornell University

Ithaca, NY, USA

Teaching Assistant — *2021 – 2022*

- CS 4740 – Natural Language Processing (Fall 2022, Lecturer: Prof. Lillian Lee).
- CS 5780 – Introduction to Machine Learning (Spring 2022).

Carnegie Mellon University

Doha, Qatar

Teaching Assistant — *2016 – 2019*

- 11-785 – Introduction to Deep Learning (Fall 2018, Spring 2019, Lecturer: Prof. Bhiksha Raj).
- 15-251 – Great Theoretical Ideas in Computer Science (Spring 2018, Spring 2019).
- 15-213 – Introduction to Computer Systems (Fall 2017).
- 15-112 – Fundamentals of Programming (Fall 2016, Fall 2017).
- 21-127 – Concepts of Mathematics (Spring 2016).